

LABTASK 1:

SOURCE CODE:

```
#include <iostream>
using namespace std;

class MedicalRecord{
private:
    string disease_name;
    double roomCharges;
    double treatmentCost;
public:
    MedicalRecord(){
        disease_name = "null";
    }
    MedicalRecord(string dn, double r, double t):
        disease_name(dn), roomCharges(r), treatmentCost(t){}

    double calculate_totalCharges(){
        return roomCharges+treatmentCost;
    }

    void display_medicalRecord(){
        cout<<"DISEASE NAME: "<<disease_name<<endl;
        cout<<"ROOM CHARGES: "<<roomCharges<<endl;
        cout<<"TREATMENT COST: "<<treatmentCost<<endl;
    }
};
```

```

class patientAdmission{
private:
    int admissionID;
    string patientName;
    MedicalRecord md1;
public:
    patientAdmission(int id, string pn, string dn, double r, double t ): md1(dn, r,t){
        admissionID=id;
        patientName=pn;
    }

    void display_patientAdmission(){

        cout<<"ADMISSION ID: "<<admissionID<<endl;
        cout<<"PATIENT NAME: "<<patientName<<endl;
        md1.display_medicalRecord();
        cout<<"TOTAL BILL: "<<md1.calculate_totalCharges()<<endl;
    }
};

```

```

int main(){
    patientAdmission ad1(1, "Ali", "Cancer", 25000, 15000);
    patientAdmission ad2(2, "Ahmed", "Pneumonia", 27000, 2000);
    patientAdmission ad3(3, "Ehtisham", "Stomach pain", 30000, 2500);

    cout<<"====PATIENT 1 DETAILS===="<<endl;
    ad1.display_patientAdmission();
    cout<<"====PATIENT 2 DETAILS===="<<endl;
    ad2.display_patientAdmission();
    cout<<"====PATIENT 3 DETAILS===="<<endl;
    ad3.display_patientAdmission();

    return 0;
}

```

OUTPUT:

```
=====PATIENT 1 DETAILS=====
ADMISSION ID: 1
PATIENT NAME: Ali
DISEASE NAME: Cancer
ROOM CHARGES: 25000
TREATMENT COST: 15000
TOTAL BILL: 40000
=====PATIENT 2 DETAILS=====
ADMISSION ID: 2
PATIENT NAME: Ahmed
DISEASE NAME: Pneumonia
ROOM CHARGES: 27000
TREATMENT COST: 2000
TOTAL BILL: 29000
```

```
TOTAL BILL: 29000
=====PATIENT 3 DETAILS=====
ADMISSION ID: 3
PATIENT NAME: Ehtisham
DISEASE NAME: Stomach pain
ROOM CHARGES: 30000
TREATMENT COST: 2500
TOTAL BILL: 32500
PS C:\Sem 2\OOP LABS>
```

LABTASK 2:

SOURCE CODE:

```
#include <iostream>
using namespace std;

class Professor{
private:
    int ProfessorID;
    string ProfessorName;
    string Specialization;
public:
    Professor(int id, string n, string s):
    | ProfessorID(id), ProfessorName(n), Specialization(s){}
    void show_professor(){
        | cout<<"PROFESSOR'S ID: "<<ProfessorID<<endl;
        | cout<<"PROFESSOR'S NAME: "<<ProfessorName<<endl;
        | cout<<"SPECIALIZATION: "<<Specialization<<endl;
        | }
};
```

```
class department{
    | string deptName;
    | Professor* pobj;
public:
    | department(string depn, Professor* p): deptName(depn), pobj(p) {}
    void show_dept(){
        |
        | | cout<<"DEPARTMENT NAME: "<<deptName<<endl;
        | | pobj->show_professor();
        | }
};
```

```

int main(){
    Professor p1(101, "JAMEEL KHAN", "ZOOLOGY");
    Professor p2(102, "SHAWAIZ SHAH", "MATHEMATICS");

    department dept("AI", &p1);

    cout<<"====DEPARTMENT ASSIGNED===="<<endl;
    dept.show_dept();
    cout<<"====DEPARTMENT NOT ASSIGNED==== "<<endl;
    p2.show_professor();
    return 0;
}

```

OUTPUT:

```

> cd "c:\Sem 2
====DEPARTMENT ASSIGNED====
DEPARTMENT NAME: AI
PROFESSOR'S ID: 101
PROFESSOR'S NAME: JAMEEL KHAN
SPECIALIZATION: ZOOLOGY
====DEPARTMENT NOT ASSIGNED====
PROFESSOR'S ID: 102
PROFESSOR'S NAME: SHAWAIZ SHAH
SPECIALIZATION: MATHEMATICS
PS C:\Sem 2\OOP LABS>

```